

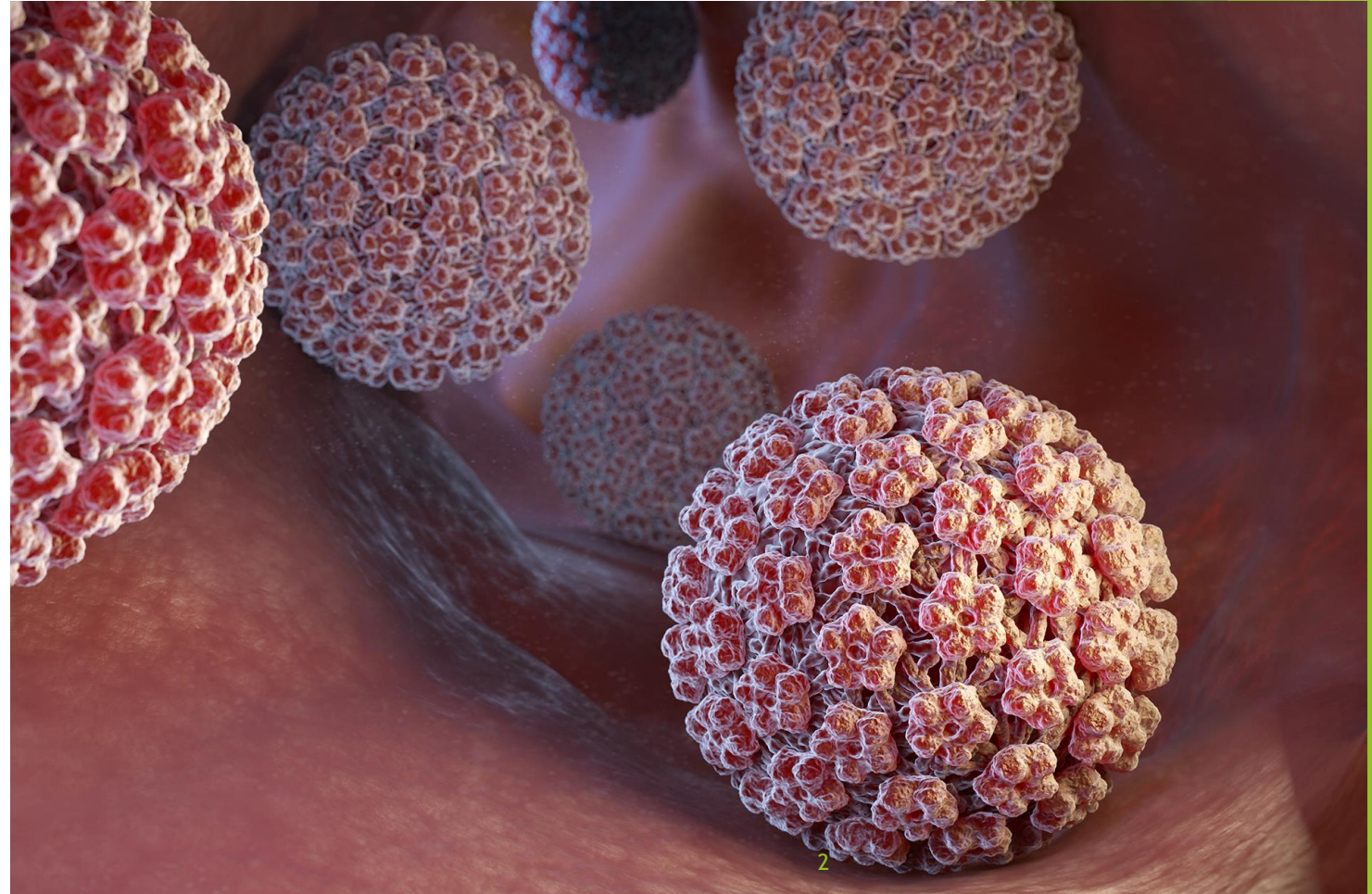
Wart parasitic infection

Dr eman gomaa

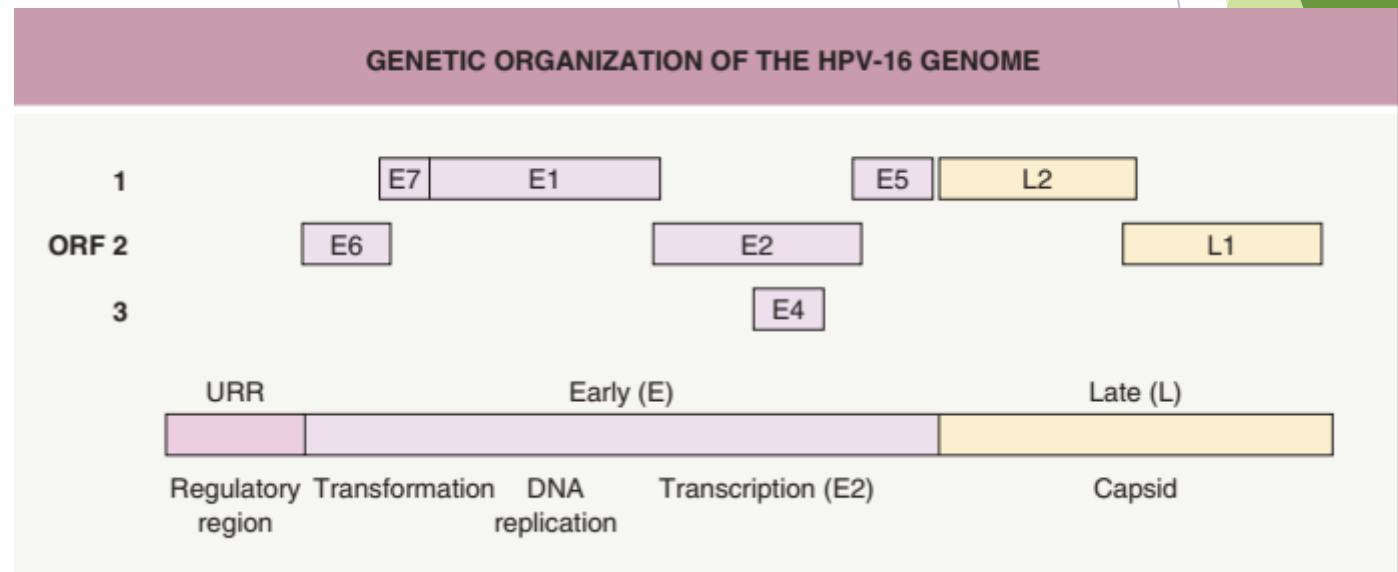
Consultant dermatologist

Human papilloma virus

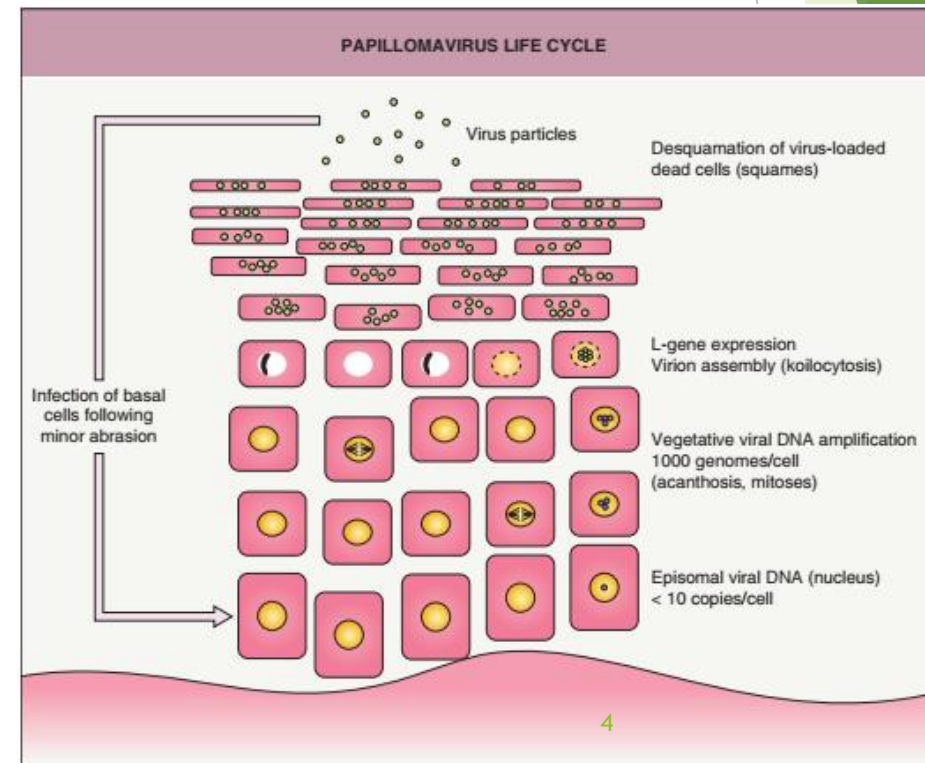
- ▶ 150 type
- ▶ DSDNA
- ▶ No envelop
- ▶ Genom :2 region:



- ▶ Early :
 - ▶ E6,7 :oncogens stimulat proliferation , prevent apoptosis
- ▶ Late :
 - ▶ L:for capsid protein
 - ▶ L1 :major capsid protein



- ▶ Epitheliotropic :skin , mucous membrane
- ▶ Life cycle :
 - ▶ Trauma →enter virus →basal cell layer →as perminant layer → autonomus replication



Oncogenic potential of HPV

- Low risk : HPV 6-11
- High risk mucosa : 16 - 18
- High risk skin : 5-8

CLINICAL MANIFESTATIONS AND ASSOCIATED HUMAN PAPILLOMAVIRUS (HPV) TYPES		
	Frequently detected	Less frequently detected
<i>Skin lesions</i>		
• Common, palmar, plantar, myrmecial and mosaic warts	1, 2, 27, 57	4, 29, 41, 60, 63, 65
• Flat warts	3, 10	28, 29
• Butcher's warts	7	1, 2, 3, 4, 10, 28
• Digital squamous cell carcinoma and Bowen disease	16	28, 31, 33, 34, 35, 51, 52, 56, 73
• Epidermodysplasia verruciformis (EV)	3, 5, 8	9, 12, 14, 15, 17, 19-25, 36-38, 47, 49, 50, etc.
• EV - squamous cell carcinoma	5, 8	14, 17, 20, 47
<i>Mucosal lesions</i>		
• Condylomata acuminata	6, 11	40, 42-44, 54, 61, 70, 72, 81
• High-grade intraepithelial neoplasias (including cervical condylomata plana, bowenoid papulosis, erythroplasia of Queyrat) and invasive cancer	16	18, 26*, 31, 33, 35, 39, 45, 51, 52, 53*, 56, 58, 59, 62, 66*, 68, 73, 82
• Buschke-Löwenstein tumor	6, 11	
• Recurrent respiratory papillomatosis, conjunctival papillomas	6, 11	
• Heck disease (focal epithelial hyperplasia)	13, 32	

wart

epidermis

Anogenital

Benign
Wart
Hpv
6-11

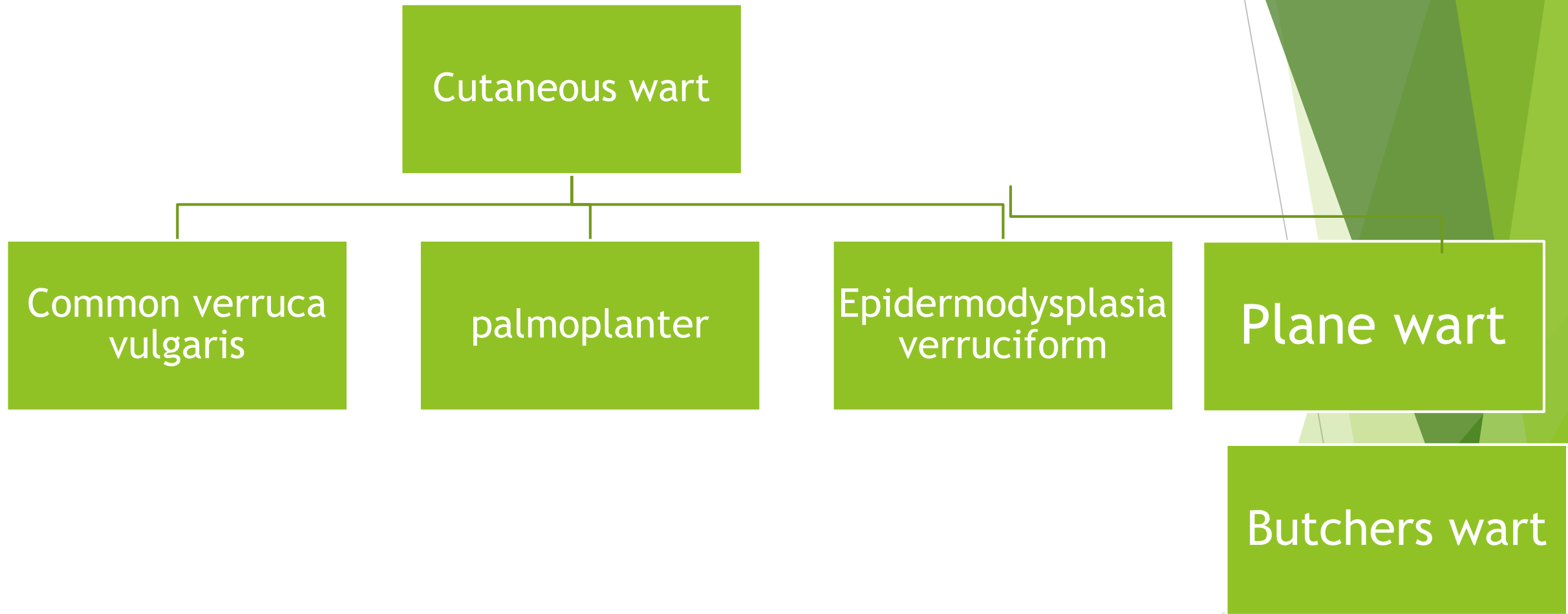
Malignant
Scc
EV
5-8

Benign
Condyloma
acummenata
6-11

Malignant
Cervical
cancer
16-18

Cutaneous wart

- ▶ Benign tumor of skin → infection by HPV
- ▶ 20-30 % of school children
- ▶ Modes of infection :
 - ▶ Direct
 - ▶ Indirect



Common verrucae vulgaris	Palmer -planter	Epidermodysplasia verruciform
Hpv 1-2-4-27-57	11-2-27-57	HPV 5-8
Asymptomatic painfull hyperkeratotic exophytic dome shaped papules • Punctate blackdots • Pseudokoebner	6-10 years Thick ,endophytic ith central depressin (ant hill) Palm , sole -beanth pressur area	Malignant wart 1. Plane wart like + 2 - TVC LIKE LESION Benign with high risk for malignancy
Fingers , site of trauma -knee, elbow	Palm , sole	
Varients : Filiform :periorofacial Periungual :nail pitting onychodystrophy	Varients : 1. mosaic :large plaque 2. Extensive chronic verricosa :immunocomprimised 3. Epidermiod planter cyst	Complication : Actinic keratosis Invasive SCC Low metastasis Increase by sun
DD:		



Fig. 79.4 Verrucae vulgares (common warts). Courtesy, A Geusau, MD.



Fig. 79.7 Multiple filiform warts on the lower face.



Fig. 79.6 Verrucae plantares (plantar warts). The photo was taken after shaving of the hyperkeratotic surface; the black dots represent hemorrhage into the stratum corneum.



Fig. 79.10 verrucae planae (flat warts). Multiple skin-colored or pink (A) to brown (B) smooth-surfaced, flat-topped papules. These lesions are typically caused by HPV-3 or -10. © Courtesy, Julie V. Schaffer, MD.





Fig. 79.11
Epidermodysplasia
verruciformis. Confluent
scaly papules and
plaques resembling flat
warts.



Fig. 79.12
Epidermodysplasia
verruciformis. Generalized
erythematous macules
and plaques are seen in
the same patient as
shown in [Fig. 79.11](#).
Lesions tested positive
for HPV-8 and -36.

DD:TB hypertrophic lp	Corn Calleus Ppk Porokeratosis Pitted keratolysis	

Plane wart	Butchers warts
3-10-28-29	Hpv 2-7
Smooth skin colored flat topped papules	Meat processing professional • Extensive verrucous papules
Face , dorsum of hand , arm	Hand , finger
DD: l.p , l.nitidus Gottron papules Acrokeratosis verruciform of hopf	

Histeopathology

- ▶ Hyperkeratosis
- ▶ Acanthosis
- ▶ Hypergranulosis , papillomatosis
- ▶ Koilocytes (hollow cells)
 - ▶ Owl eye appearance
- ▶ Church spire
- ▶ Thrombosed vessles

Anigenital wart condyloma acumenata

- ▶ Low risk :6-11
- ▶ High risk :16-18
- ▶ Mode of transmtion :sexual , fingers , underwear
- ▶ Young adult male >female

c/p

1. Condyloma acuminata: classic
 1. Cauliflower like
2. Papular
3. Flat topped papules
 1. Condyloma plana
4. Keratotic wart



Fig. 79.13 Condylomata acuminata. Verrucous lesions on the glans and in the sulcus, with a few small papules on the shaft of the penis and distal glans. Note the acuminate topography, i.e. the tapering to a point. Courtesy, Lorenzo Cerroni, MD.



Fig. 79.14 Condylomata acuminata. A Confluent lesions forming hyperpigmented plaques in the perineal region and along the inguinal fold. A depigmented scar is seen at the site of prior treatment with liquid nitrogen. B Large, exophytic, broad-based or pedunculated papillomas in a healthy 15-year-old boy.

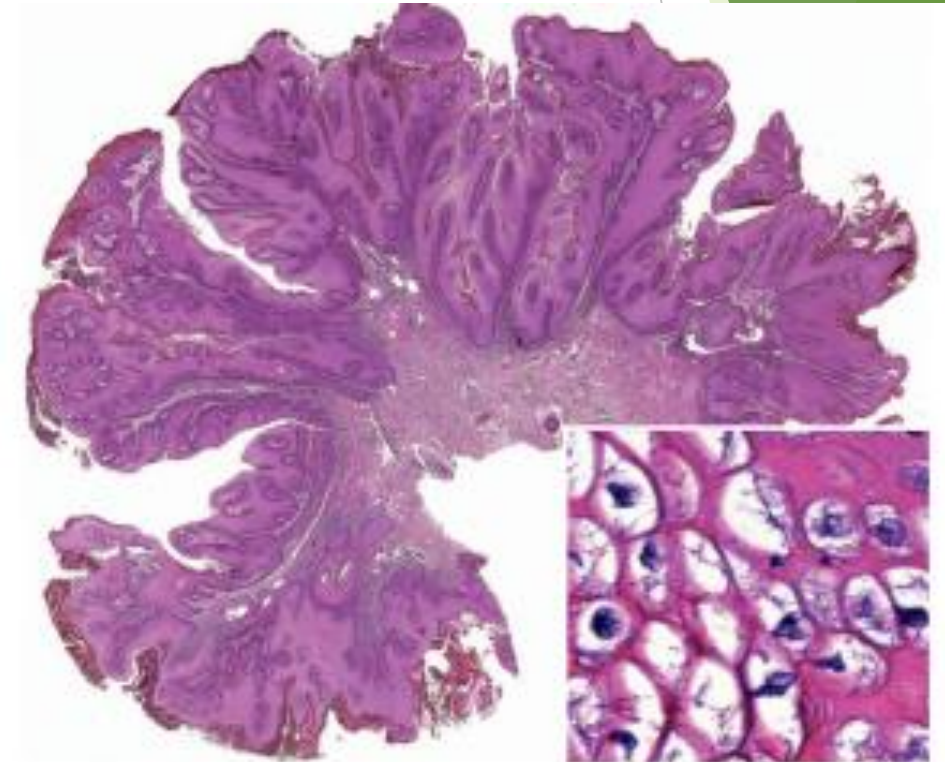


Fig. 79.19 Buschke-Löwenstein tumor – histopathologic features. Exophytic lesion with a papillomatous surface and marked, irregular epithelial hyperplasia without cellular atypia as well as the characteristic downward extension of bulbous rete ridges. Note vacuolated keratinocytes (inset). Courtesy, Luk

Clinical variants (8)

1-Buschkelowenstein

Gaint condyloma acumenata

Low risk 6-11

Cauliflower → fistula → abscess

Benign → mild malignancy

Verrucous carcinoma



Bowenoid

Brown plaque

HPV 16-18

Sc16-18

SCC insitu



positive for high-risk mucosal HPV in a man who had sex with men. Histology revealed high-grade anal intraepithelial neoplasia (AIN), not invasive SCC. However, continued monitoring is indicated

Erythroplasia of querat

Velvety erythematous plaque on glans penis

Hpv 16

SCC insitue



Fig. 79.17 Erythroplasia of Queyrat. A well-demarcated velvety plaque of the prepuce positive for high-risk HPV; histology revealed high-grade penile intraepithelial neoplasia (PIN).

Hecks

Focal epithelial
hyperplasia
Flat wart at gingiva ,
buccal , labial
Hpv 13 -32

Oral

6-11
Digital , oral sex
Leukoplacia :hpv 16
Oropharengial
carcinoma

Conjunctival

Hpv6-11

Nasal

6-11-16-18
scc

Respiratory
6-11



Fig. 79.20 Oral warts. Papillomas of the labial mucosa in a 6-year-old child.

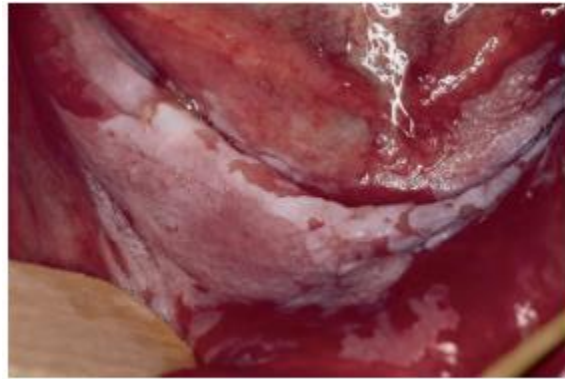


Fig. 79.25 Verrucous proliferative leukoplakia of the oral cavity. The lesions have focally progressed to invasive squamous cell carcinoma; HPV DNA was

TREATMENT OF ANOGENITAL WARTS

Clinician and patient decide on treatment based on morphology and distribution of lesions and suitability of treatment plans

Patient-applied therapy

- Podophyllotoxin
- Imiquimod
- Sinecatechins

Clinician-applied therapy

- Cryotherapy
- Trichloroacetic acid (TCA) 80%–90%
- Removal by scissor or shave excision, curettage, electrosurgery, laser vaporization, or excisional surgery

Special situations

- Large or disseminated intraepithelial neoplasias
- Immunosuppression
- Buschke–Löwenstein tumor
- Pregnancy
- Children

Refer to specialist

Treatment of wart

Local destructive method

Topical , intralesional
cytotoxic therapy

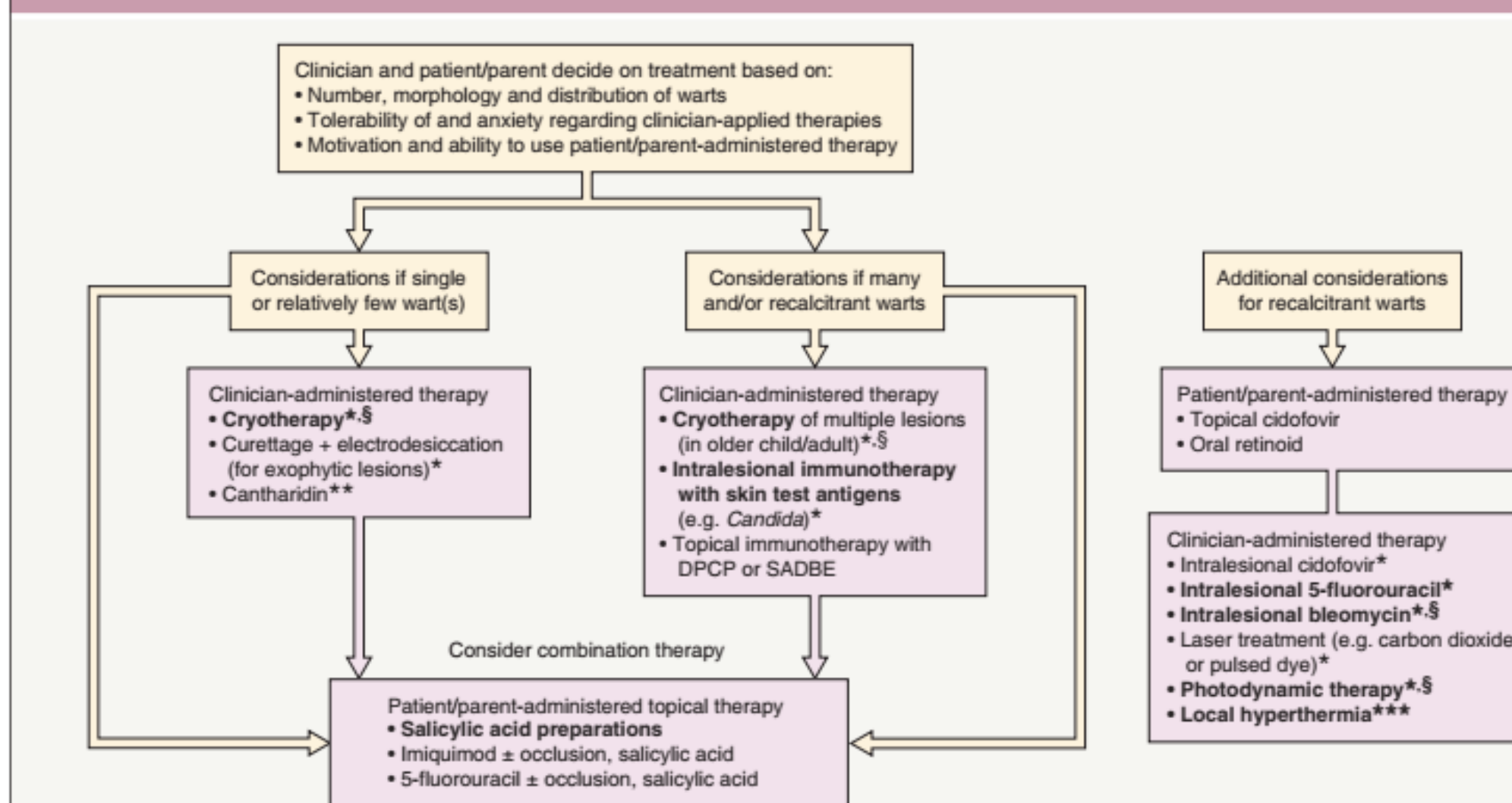
Topical
immunomodulator

Systemic
immunomodulator

Immunotherapy

Special situation

TREATMENT OF COMMON WARTS



1)Local destructive method

Local destructive method

1. Cryotherapy :
2. TCA70-90%
3. Curatage
4. Electro
5. Laser tt: ablative ,vascular
6. Photo dynamic therapy

Topical ,intralesional cytotoxic therapy

- ▶ Podophylin
- ▶ Podphylotoxin
- ▶ Flurouracil
- ▶ Pleomycin

Topical immunomodulator

- ▶ Imiquimod 5%

Systemic immunomodulator

- ▶ Interferon
- ▶ Oral cimetidine
- ▶ Oral zinc
- ▶ Gardasil
- ▶ Cervarix

Immunotherapy

- ▶ Candida: 0,3 ml every 3 weeks
- ▶ MMR : 0,3 ml every 3 weeks → 5 times
- ▶ PPD : 0,1 ml

Molluscum contagiosum

- ▶ Age :
 - ▶ Children :direct contact
 - ▶ Adult :sexualy
 - ▶ Causitive organisim :2 type
 - ▶ Mcv1 :majorty
 - ▶ Mcv2 :adult , sexual



c/p

- ▶ Umbilicated chiny pearly white papules
- ▶ Site :anywhere ,axilla , neck
- ▶ Facial , anogenital



Varients

- ▶ Agminate
- ▶ Wide spread gaint → HIV
- ▶ Oral
- ▶ FOLLICULAR
- ▶ Scar

Histeopathology

- ▶ Acanthosis
- ▶ Inclusion body : molluscum body
 - ▶ Handerson -petterson body
 - ▶ Larg eosinophilic , intracytoplasmic
- ▶ Central crater (epidermal invagination)
- ▶ No inflammation , infiltrate

- ▶ **DD:**
 - ▶ Plan wart ,
 - ▶ pyogenic granuloma
 - ▶ Spitz nevus
- ▶ **Course :**self limited in several months , several years

► TT :

- Therapy is not required
- TCA , phenol , salicylic
- Retinoid
- Surgical , cryo
- Immunological therapy

TREATMENT OF MOLLUSCUM CONTAGIOSUM 
No therapy
Eventual spontaneous involution in immunocompetent patients Risks of spread (especially in atopic patients), associated dermatitis, and pruritus
Physical modalities
Cryotherapy (2) Curettage (1) Electrodesiccation (3) Manual extraction (3)
Topical and intralesional therapy
Cantharidin (2*) Cidofovir (2) Podophyllotoxin (1) Potassium hydroxide 5–10% solution (1) Salicylic and lactic acids (2) Retinoids (3) Silver nitrate paste (2) Trichloroacetic acid (1) Candida antigen, intralesional (2) Interferon- α , intralesional (3)
Laser therapy
CO ₂ (3) Pulsed dye (2)
Systemic therapy
Cidofovir (in HIV-infected patients) (3) Cimetidine (2) Griseofulvin (3)

Poxvirus infection

ORF

- ▶ Ecthyma contagiosum
- ▶ Parapox virus
- ▶ Direct contact with infected animals

Molker nodules

- ▶ Paravaccinia , pseudocowpox
- ▶ Paravaccinia virus
- ▶ Milkers , teats
- ▶ I.P :4 days, weeks
- ▶ Finger , hand , arm
- ▶ Self limited
- ▶ Mammary glands of animals

SELECTED POXVIRUS INFECTIONS					
Disease	Virus/genus	Host	Clinical findings	Treatment	Comment
Monkeypox	Monkeypox/ Orthopoxvirus	Monkeys (especially captive), rodents, humans	- Malaise, fever and flu-like symptoms precede eruption - Number of lesions varies from a few to >100 - Papules progress to vesiculopustules then crust - Various stages may be present simultaneously - Usually has a centrifugal distribution⁴¹; often involving the palms and soles - Lymphadenopathy and respiratory symptoms common	Supportive; post-exposure smallpox vaccination; cidofovir for severe cases; usually self-limited course; lesions heal with scarring	- Primarily in Africa - Acquired via cutaneous inoculation or inhalation; human-to-human may be possible^{40,41} - Incubation period = ~10–12 days - Outbreak in US in 2003 due to human contact with infected prairie dogs exposed to an infected Gambian giant rat from Ghana⁴⁰ - Laboratory diagnosis: variola real-time PCR (–), orthopox real-time PCR (+), viral culture; acceptable specimens as for smallpox - Smallpox vaccination protective
Cowpox	Cowpox/ Orthopoxvirus	Cattle (now a rare host), cats, rodents, humans	- Painful papule at site of contact with lesions on infected animals (usually cats) - Hands and face most common sites of involvement - Erythematous, edematous papule rapidly becomes vesicular, pustular, then hemorrhagic - Progresses to ulcer with crust and eschar - Lymphadenopathy, fever and flu-like symptoms	Supportive; heals over 4 weeks with scarring; scars often large and depressed	- Primarily in Europe and Central Asia - Less common than orf or milker's nodule - May be transmitted by cats with subclinical infections - Incubation period = 7 days - Occasionally severe disease, including generalized eruptions in patients with atopic dermatitis
Orf (ecthyma contagiosum, sore mouth disease, scabby mouth disease, contagious pustular dermatitis, infectious pustular dermatitis)	Orf/ Parapoxvirus	Sheep, goats, reindeer, humans	- Papules (1 to few) on hands (Fig. 81.12A) at sites of contact with infected animals (lesions favor perioral area and udders of ewes), carcasses, or fomites (e.g. fences, barn doors) - Progression through several stages: - maculopapular - targetoid - weeping nodule - regenerative dry stage with black dots - papillomatosis - regression with a dry crust - May be accompanied by lymphangitis, lymphadenopathy, malaise, fever	Supportive; topical imiquimod may stimulate early regression; usually heals without scarring	- Endemic in sheep and goats - At-risk occupations: shepherds, butchers, veterinary surgeons, sheepshearers - Histologic findings depend on stage of the lesion and include epidermal necrosis, vacuolated keratinocytes, a dense mixed dermal infiltrate, and delicate finger-like projections of the epidermis into the dermis; eosinophilic inclusion bodies (intracytoplasmic > intranuclear) (Fig. 81.12B) - Laboratory diagnosis: histologic examination of skin lesion ± EM; parapox real-time PCR (+), orf real-time PCR (+) - Erythema multiforme occasionally occurs 10 to 14 days later
Milker's nodules (bovine papular stomatitis, pseudocowpox)	Paravaccinia/ Parapoxvirus	Cattle, humans	- Papule develops at site of contact with lesions on infected animals (e.g. muzzles of calves, teats of cows) - Often single lesions; favor hands and forearms - Cutaneous lesions virtually identical to orf	Supportive	- Endemic in cattle - At-risk occupations: dairy farmers, ranchers, butchers, veterinary surgeons - Laboratory diagnosis: histologic findings ± EM (see orf); parapox real-time PCR (+), paravaccinia real-time PCR (+)
Deer-associated parapoxvirus infection	Parapoxvirus	Deer, humans	- Friable violaceous nodule at site of non-healing cut obtained while field dressing a deer carcass - Single lesion on a finger - Fever following biopsy in one patient	Supportive	- Reported in deer hunters in the Eastern US⁴² - Histologic findings include epidermal hyperplasia ± necrosis, intracytoplasmic eosinophilic inclusion bodies in keratinocytes, dilated vascular spaces with swollen endothelial cells, and variable lymphohistiocytic and neutrophilic infiltrates - Parapox real-time PCR (+)
Tanapox	Tanapox/ Yatapoxvirus	Non- human mammals	Isolated skin lesion, usually on the leg	Supportive; heals over 6–8 weeks with scarring	- Endemic in equatorial Africa - Transmission to humans probably occurs via an infected vector, although direct

COMPARISON OF VARICELLA/DISSEMINATED ZOSTER TO SMALLPOX		
	Varicella/disseminated zoster	Smallpox
Prodrome	None or mild fever and malaise	Fever $\geq 101^{\circ}\text{F}$ (38.3°C) for 1 to 4 days prior to onset of eruption, plus prostration, headache, chills, backache, and/or gastrointestinal symptoms
Distribution of lesions	Face/scalp initially; trunk affected more than distal extremities May have an enanthem	Lesions concentrated on face and limbs; can progress to involve entire body surface Enanthem often precedes cutaneous eruption
Stage of lesions	Different stages (papules, vesicles, pustules) are seen simultaneously in any one area of the skin	Adjacent lesions are all at same stage of development (synchronous)
Type of lesions	Relatively superficial, with evolution from macules/papules to vesicles to pustules	Begin as papulovesicles and then become firm, deep-seated pustules with a tendency to confluence
Course	Lesions tend to appear in crops over a 3-day period	Lesions spread over 1–2 weeks, and crusting develops over 1 week
Scarring	Rare in uncomplicated cases	Common and marked

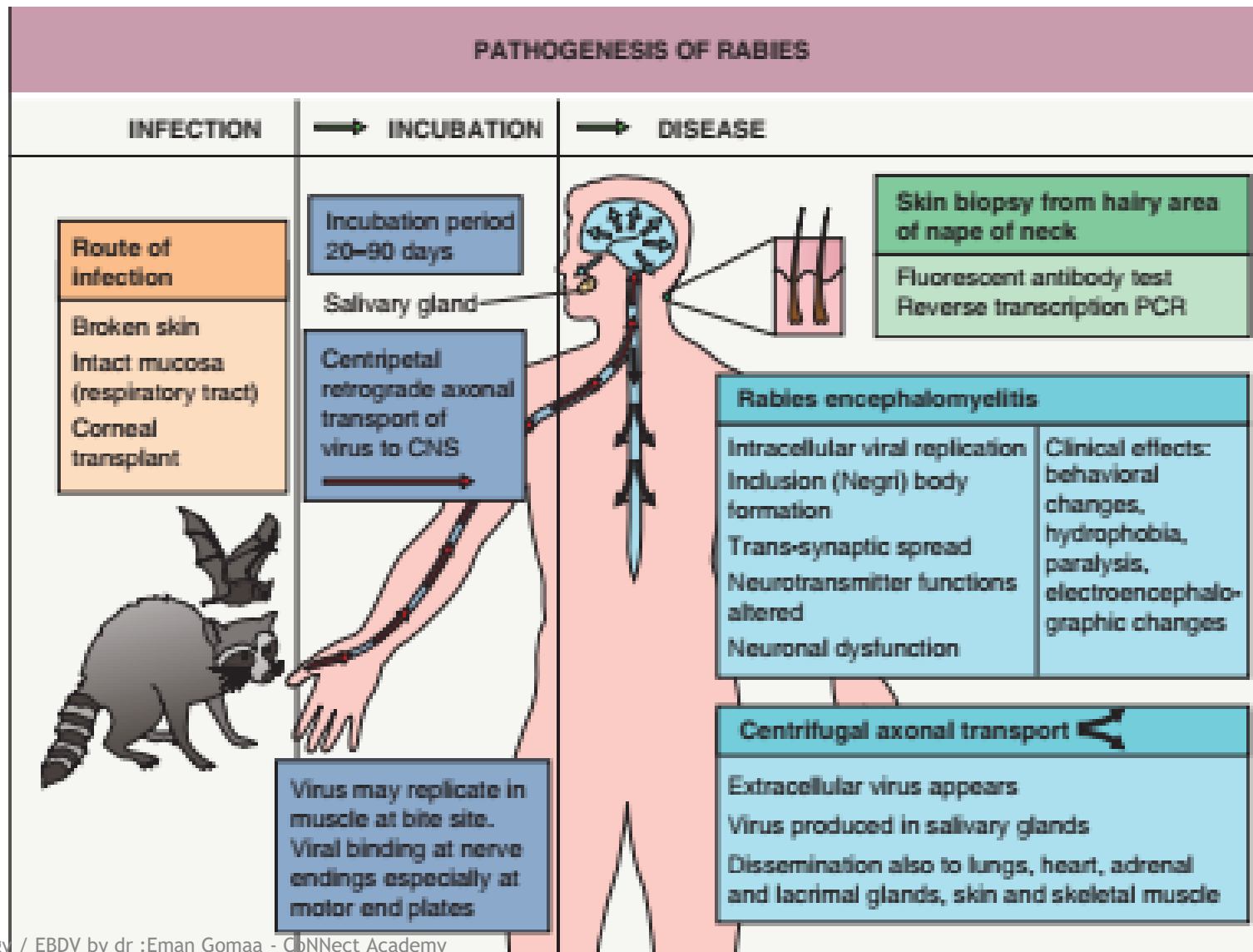
measles	Rubella	Erythema infectiosum	Gianoti crosti	HFMD

CUTANEOUS MANIFESTATIONS OF HEPATITIS B AND/OR C INFECTION

Small vessel vasculitis (B, C)
Cryoglobulinemic vasculitis (C>B)
Urticarial vasculitis (B, C)
Polyarteritis nodosa (B [classic]>C)
Livedo reticularis (C)
Serum sickness-like reaction (B, C)
Urticaria (B, C)
Gianotti–Crosti syndrome (B>C)

Necrolytic acral erythema (C)
Porphyria cutanea tarda (B, C)
Pruritus (B, C)
Lichen planus – particularly erosive oral disease (C)
Sarcoidosis (with interferon and/or ribavirin therapy^{*}; C>B)
Erythema multiforme (B, C)
Erythema nodosum (B>C)

^{*}Employed less nowadays.



Parasitic infection

Parasitic cutaneous infestation

Arthropods

Insecta
6 legs
Mousquito ,flies
pediculosis

Arachnida
8 legs
Spider ticks ,
mite
Scabies

Protozoa
Single cell
Leishmania

Helmensis
Cutaneous larva migrans

leishmaniasis

Leishmania

L donovani

L chagasi
L infantum
L aethiopica
L tropica
L denovani
L major

L Mexicana
L mexincana
L amazonensis
L venezuelensis

L braziliensis
L brazile
L panamensis

EPIDEMIOLOGIC AND CLINICAL PATTERNS OF INFECTIONS WITH *LEISHMANIA* SPECIES

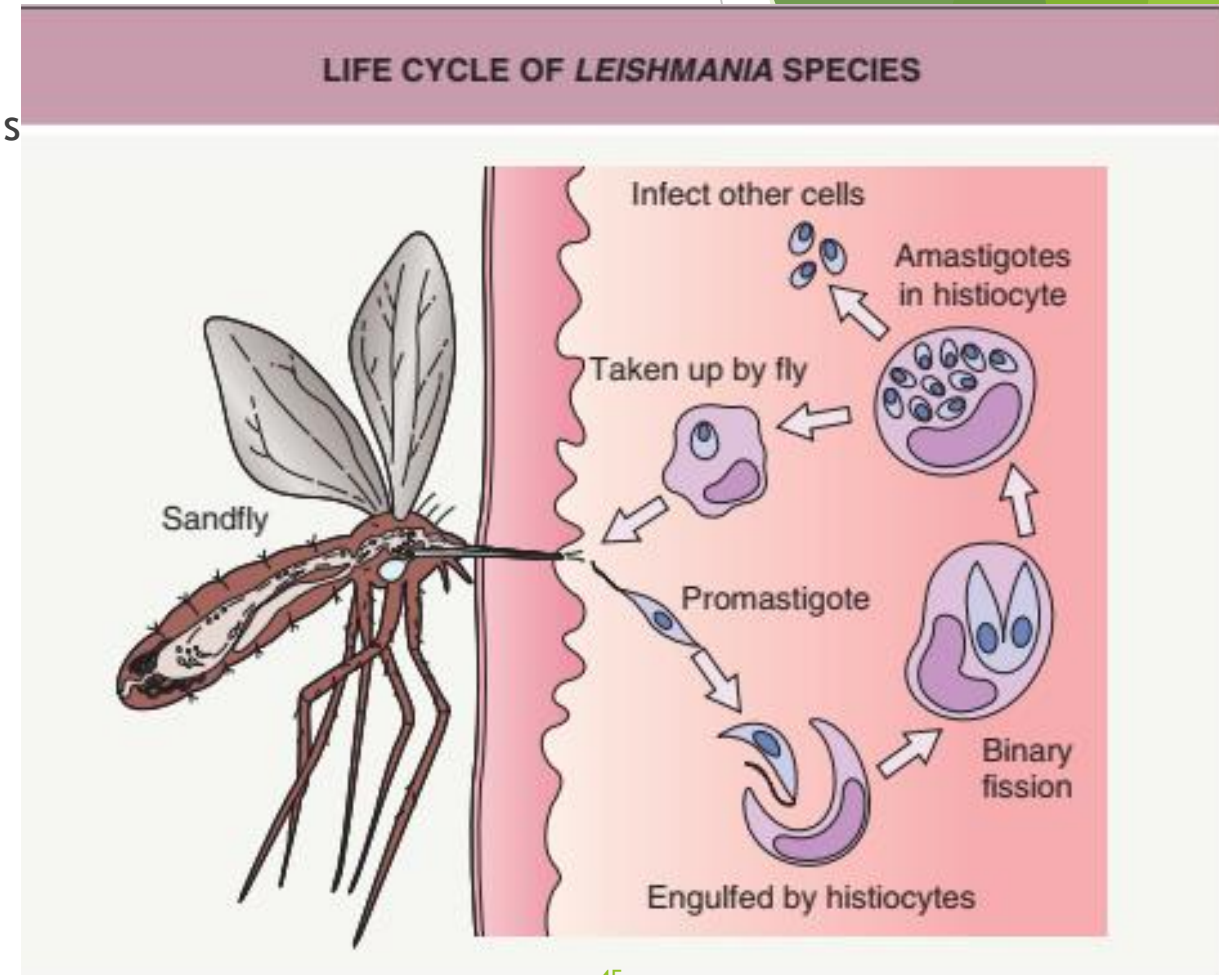
Complex	Major species	Major reservoirs	Major geographic distribution	Clinical patterns (in addition to cutaneous)	Clinical forms of the cutaneous pattern
<i>Leishmania donovani</i>	<i>L. donovani</i>	Humans	Sudan, Kenya, Tanzania, India, Bangladesh	Visceral, PKD, mucosal (rare)	(Uncommon)
	<i>L. infantum</i>	Dogs, foxes	Mediterranean region	Visceral (favors children), PKD [*] , mucosal [†] (rare)	Favors children (uncommon)
	<i>L. infantum chagasi</i> ^{**}	Dogs, foxes, opossums	Central and South America		
<i>Leishmania tropica</i>	<i>L. major</i>	Rodents	Arid areas of Africa (north and south of the Sahara), Middle East and Central Asia	Mucosal (rare)	Old World zoonotic/rural (moist, early ulcerative with rapid course), recidivans (rare), lupoid (rare)
	<i>L. tropica</i>	Humans, dogs	Eastern Mediterranean region, Middle East, Central Asia	Visceral (rare), mucosal (rare)	Old World anthroponotic/urban (dry, late ulcerative with chronic course), recidivans, lupoid
	<i>L. aethiopica</i>	Hyraxes	Ethiopia, Kenya, Yemen	Diffuse cutaneous, mucocutaneous	Dry/late ulcerative or moist/early ulcerative
<i>Leishmania mexicana</i>	<i>L. mexicana</i>	Forest rodents	Mexico, Central America	Diffuse cutaneous (rare)	New World Variants include ulcerative, plaque-like, pustular, impetigo-like, erysipeloid, sarcoidosis-like, papulotuberculous, verrucous, sporotrichoid, eczematous
	<i>L. amazonensis</i>	Forest rodents	South America	Diffuse cutaneous, mucocutaneous, visceral (rare), PKD (rare)	
	<i>L. venezuelensis, pifanoi, gamhami</i>	Unknown	Venezuela	Diffuse cutaneous (rare)	
<i>Leishmania (Viannia) braziliensis</i>	<i>L. braziliensis</i>	Forest rodents	Central and South America	Mucocutaneous, visceral [†]	
	<i>L. guyanensis</i>	Sloths, anteaters	Guyana, Brazil	Mucocutaneous	
	<i>L. panamensis</i>	Sloths	Panama, Costa Rica, Colombia	Mucocutaneous	
	<i>L. peruviana</i>	Dogs	Peru, Argentina		

- ▶ Reservoir :animals
 - ▶ Canines, (dogs , cat , rodents)
- ▶ Vector :
- ▶ Infected femal sand fly
 - ▶ Phlebotomus →old
 - ▶ Lutozomyia →new
- ▶ Humen→accidental host
- ▶ No humen to humen
- ▶ Except →l tropica

► Life cycle :

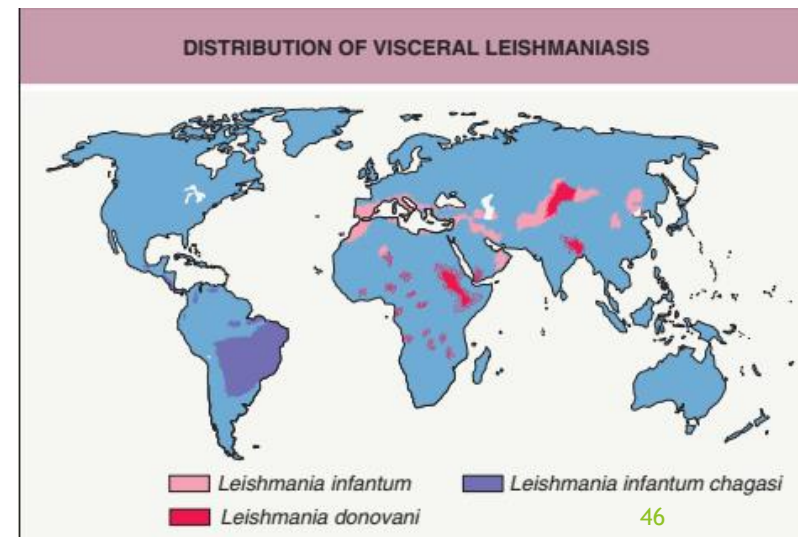
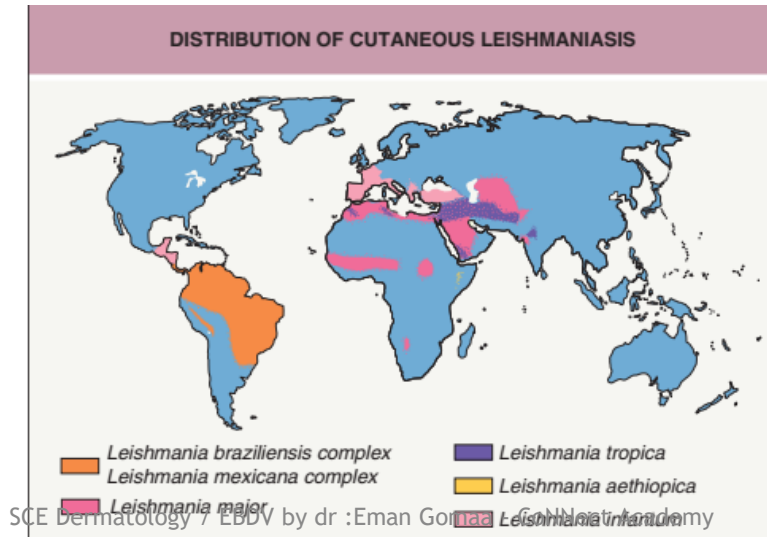
- Promastigote → infective stage → extracellular in gut of sandfly
- Spindle , flagellated
- Engulfed → amastigote intracellular in histiocyte

COUNTRIES WITH THE HIGHEST INCIDENCE OF LEISHMANIASIS	
Old World cutaneous leishmaniasis	Afghanistan, Algeria, Iran, Iraq, Saudi Arabia, Syria
New World cutaneous leishmaniasis	Brazil, Peru, Colombia
Mucocutaneous/mucosal leishmaniasis	Bolivia, Brazil, Peru
Visceral leishmaniasis	Brazil, Ethiopia, India, Kenya, Somalia, South Sudan, Sudan.



Fare of infection

- ▶ Most don't result clinical disease
- ▶ Minor → clinical cutan- leishmaniasis
 - ▶ Th1:il2, IFN gamma → quick resolution
 - ▶ Th2 → il4 -il10 → increase disease
- ▶ Epidemiology :
 - ▶ Brazil ,iran , afghanistan , sudan south America , asia , Africa ,Egypt



Clinical feature

Cutaneuse

Mucocutaneus

visceral

Acute localized

Chronic lupoid

Diffuse anergic ,
lepomatous

Recurrent recidivans

Cutaneous

Acute localized	Chronic lupoid	Diffuse anergic (lepromatous)	Recurrence (recidivans)
Papule → nodule, plaque → ulcerat , verrucous Solitary or multiple Sporotrichoid pattern	Anthroponotic Few amastigote Difficult to find diagnosis Lupus vulgaris like	L athiopica L amazonsis Mainly new world Multiple keloid like lesion Face , limp	Recurrence of original Unhealthy
Exposed site → ear face , neck , arm , leg	h.p: epithelioid granuloma +lymphocyte	DD :LEPROSY : But no destruction of nasal septum	
Old world 1. rural :l major , zoonotic , moist , mild early ulcer , 2. Urban :l tropica, anthrobonotic , dry , late , chronic			
New world :like eczema , impetigo , sarcoidosis, erysiploid			



Fig. 83.4 Variable presentations of cutaneous leishmaniasis. Ulcerated plaques with rolled border (**A**) and central crusting (**A, B**). Plaque with translucent borders containing telangiectasias and central scarring (**C**). Cutaneous leishmaniasis is sometimes mistaken for a basal cell carcinoma.
A, C, Courtesy, Julie V Schaffer, MD.



Fig. 83.8 Diffuse and disseminated cutaneous leishmaniasis. **A** Diffuse cutaneous leishmaniasis presenting with large nodules that resemble lepromatous leprosy. **B** Small papules in disseminated cutaneous leishmaniasis.

Investigaion

- ▶ H.P:- diffuse suppurative granuloma, Amastigote, denovani [gemma stain].
- ▶ Culture nicolle novy macNeal.
- ▶ Leishman reaction [montenegro] → cell mediated response.
- ▶ Smear → edge of new ulcer
- ▶ PCR.

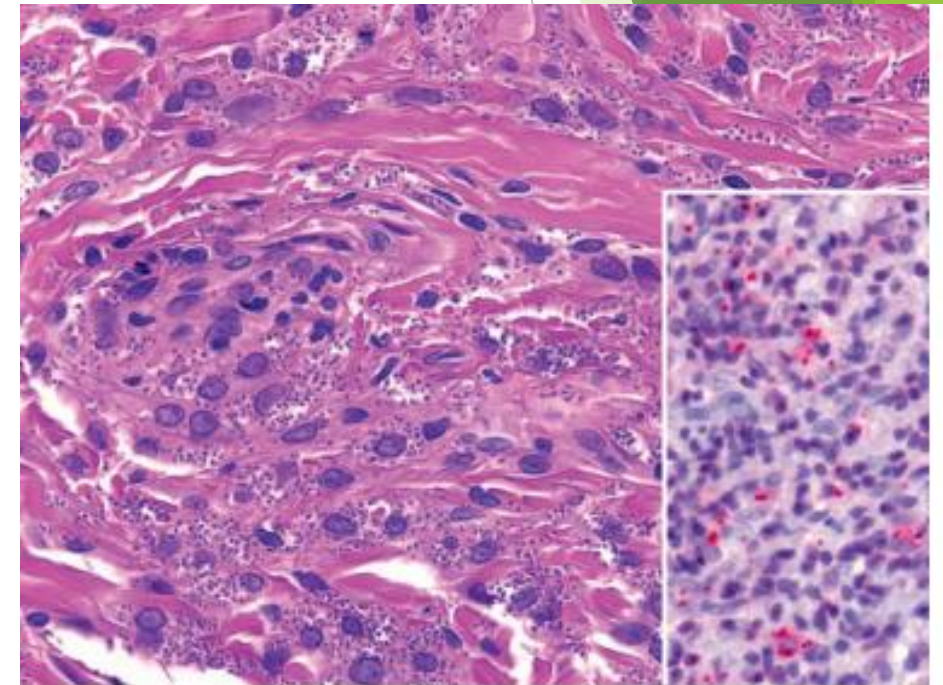


Fig. 83.12 Histopathology of leishmaniasis. Parasitized macrophages that contain amastigotes are seen in the dermis and highlighted by CD1a immunostaining (inset). Each amastigote has a nucleus and kinetoplast (not

Mucocutaneous

- ▶ In new world only.
- ▶ L-brazilience.
- ▶ I.P:- months → 20 years.
- ▶ Edema at lips & nose → infiltration, ulceration & perforation.
- ▶ Destruction of the vocal cords → change phonation.
- ▶ Not heal spontaneous :- topirface → espundia.



Fig. 83.9 Mucocutaneous leishmaniasis. Infiltration of the nose and upper lip is evident. Note the destruction of the nasal alae and crusting of the upper lip.
Courtesy, Jaqueline Luque, MD.

Visceral

- ▶ Kala-azar, dum dum fere.
- ▶ Mainly old world.
- ▶ L-infantum.
- ▶ L-changasi.
- ▶ L-donovani.
- ▶ Bone marrow, spleen & liver →
- ▶ fever, cough, epistaxis, diarrhea & HSM.
- ❑ Post-kala-azar
 - ▶ → hypopigmentation macule → **black fever**.
 - ▶ Malar erythema.
 - ▶ Verrucous papule.



Fig. 83.11 Post-kala-azar dermal leishmaniasis. Nodules of various sizes, some pedunculated, are seen in this patient who had been treated for kala-azar over a period of 6 months, 20 years previously. With permission from Peters W, Pasvol G. Tropical Medicine and Parasitology, 6th edition. London: Mosby, 2007.

EPIDEMIOLOGY AND CLINICAL PATTERNS OF VISCERAL LEISHMANIASIS

Type of disease	Major species	Major geographic distribution
Kala-azar (60% 10–20 years of age)	<i>Leishmania donovani</i>	India, China (north of Yangtze River), Kenya, Sudan
Infantile kala-azar (90% <10 years of age)	<i>L. infantum</i>	Central Asia, China, Iraq, France, Italy, Egypt, elsewhere in the Mediterranean basin, Saudi Arabia
	<i>L. infantum chagasi</i>	Brazil, Colombia, Venezuela, Central America, Mexico
Oronasal kala-azar	<i>L. infantum</i> *	France
Post-kala-azar dermal leishmaniasis	<i>L. donovani</i> <i>L. infantum</i> † <i>L. amazonensis</i>	India, China, Kenya Mediterranean basin, South America
Atypical kala-azar complex (typically in adults)	<i>L. tropica</i> <i>L. amazonensis</i> <i>L. braziliensis</i> *	Northeast Saudi Arabia South America South and Central America

SYSTEMIC FEATURES OF VISCERAL LEISHMANIASIS (<i>L. DONOVANI</i>)	
Clinical features	Proportion affected (%)
Age <9 years	20 (<i>L. infantum</i> and <i>L. chagasi</i> more commonly affect children and infants)
Age <15 years	45
Fever	85–100
Wasting	70–100
Loss of appetite	60–75
Cough	70–85
Epistaxis	45–55
Diarrhea	25–55
Vomiting	2–35
Splenomegaly	90 (adults); >95 (children)
Hepatomegaly	55–65
Lymphadenopathy	55–85 (uncommon outside of Africa)
Jaundice	2–7
Edema	2–7
Laboratory findings	
Globulin >30 g/l	95
Albumin <30 g/l	90
Anemia	60–90
Leukopenia	85
Thrombocytopenia	75
Elevated bilirubin	15
Elevated liver transaminases	20
Elevated alkaline phosphatase	40
Positive <i>Leishmania</i> serology	95
Parasitologically proven	95

Treatment

Topical

Cryo.

Intra lesion :- Na - stibogluconate [pentostam].

Photodynamic therapy.

Heating.

Systemic

Na- stibogluconate.

Pentostam 100 mg/ml. → 20mg /kg/day

Others:- azole, allopurinol, itraconazole & fluconazole.

Meglumine 85 mg/ml (20 mg/Kg)

TREATMENT RECOMMENDATIONS FOR LEISHMANIASIS

Cutaneous leishmaniasis

Drugs of choice	Sodium stibogluconate*	20 mg/kg/day IV or IM x 20 days
	Meglumine antimonate*	Various intralesional regimens, e.g. 0.2–5 ml (5 sites/lesion intradermally for 0.1 ml/cm ² until blanched) every 3–21 days for 5–8 sessions or until healing
	Miltefosine**	50 mg BID if weight is 30–44 kg and TID if weight is ≥45 kg x 28 days
Alternatives	Paromomycin 15% and MBCL 12% ointment	Topically BID x 10 days, wait 10 days, then repeat
	Paromomycin 15% and gentamicin 0.5% cream	Topically daily for 20 days
	Pentamidine	3–4 mg/kg IV or IM every other day x 3–4 doses
	Fluconazole†	200 mg/day or 6–8 mg/kg/day PO x 6 weeks
	Liposomal amphotericin B†	3 mg/kg/day IV on days 1–5 + 10 or days 1–7
	Amphotericin B deoxycholate	0.5–1 mg/kg IV daily or every other day for a total dose of 15–30 mg/kg

Mucocutaneous leishmaniasis

Drugs of choice	Sodium stibogluconate*	20 mg/kg/day IV or IM x 28 days
	Meglumine antimonate*	
	Liposomal amphotericin B	~3 mg/kg/day IV for a total dose of ~20–60 mg/kg
	Amphotericin B deoxycholate	0.5–1 mg/kg IV daily or every other day for a total dose of ~20–45 mg/kg
	Miltefosine**	50 mg BID if weight is 30–44 kg and TID if weight is ≥45 kg x 28 days
Alternative	Pentamidine	2–4 mg/kg IV or IM every other day or 3 times per week for ≥15 doses

Pediculosis

Capitis

Corporis

Pthirus pubis



Fig. 84.7 Head louse family. From left to right: female, male and nymph. With

Capitis

- ▶ Epidem:- worldwide [3-11 ys. more common in girls].
- ▶ Race:- uncommon in Africa.
- ▶ Causative:- pediculosis humanus capitis → Direct head-head Or indirect → poverty, poor, hygiene and overcrowdness.
- ▶ Life cycle:- lays eggs on 5-10 days & hatch 8 days ; cement substance.

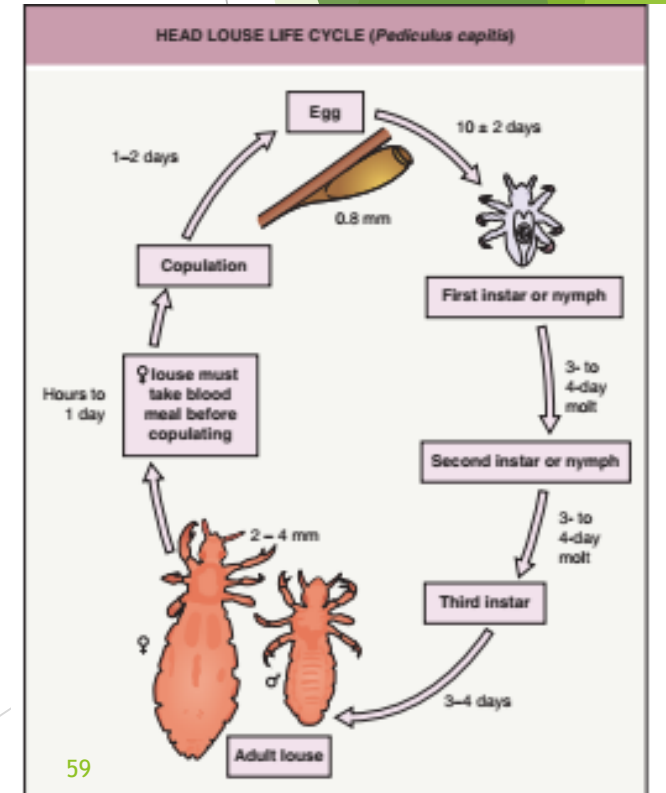


Fig. 84.8 Head louse life cycle (*Pediculus capitis*).

- ▶ C.pic:-Symptoms→ scalp pruritis :- secretions & excretion of lice → allergy & hypersensitivity reaction which takes 2-6 weeks to appear.
- ▶ Signs:- scalp behind ear, nape → excoriation, nits, lice
- ▶ & generalized pruritis.

□ Complications:-

1. Impetigo → Most common.
2. Plicapolonica.
3. Transmission of disease.



Fig. 84.9 Head lice. A The head louse egg or nit is 0.8 mm in length. B Head lice nits on hair. With permission from Espin D, Meinking TL. Infections. In: Schachner LA, Harman RC (eds). Pediatric Dermatology, 4th edn. Edinburgh: Mosby, 2011:1141-80.

Corporis

- ▶ Infestation of human clothes by body lice.
 - ▶ Pediculosis humanus humanus.
 - ▶ Nits, lice found along clothing.
 - ▶ Pruritis, back, neck, shoulder & waist area.
 - ▶ Vagabonds, morbus errorum.
- ❑ Complications.
 - ▶ Impetigo - lymphadenopathy.
 - ▶ Epidemic typhus (rickettsia).
 - ▶ Relapsing fever → (borrelia recurrent).
 - ▶ Trench fever (partonella quitana)
 - ❑ Ttt
 - ❑ Disinsection of clothes, ivermectin & topical insecticide.



Fig. 84.12 Body lice eggs in the seams of clothing.

Pthirus pubis

- ▶ Crab lice.
- ▶ STD / contact.
- ▶ Wider, shorter.
- ▶ Legs like tiny crabs.
- ▶ Male → 15-40 ys.
- ▶ Ambulate on all body surface.
- ▶ Macula caerulea → gray blue macules.
- ▶ ttt → shaving, wash clothes & eyelashes :- petrolatum asphyxia.
- ▶ In children → child abuse.



Fig. 84.10 Adult crab louse. Respiratory spiracles and circulatory system are easily seen (45x). Courtesy, Tony Burns, MD.



Fig. 84.11 Crab lice. A Both adult crab lice and nits are evident on pubic hairs. B Crab lice nits and feces on the eyelashes. Courtesy, Louis A. Fraga, Jr., MD, with permission from Tapak Meinking TL. Infestations. In: Schachner LA, Hansen RC (eds). Pediatric Dermatology, 4th ed. Edinburgh: Mosby, 2011:1743.

TREATMENTS FOR HEAD LICE				
Treatment	Group	Administration on days 1 and 8	Concerns	Efficacy & resistance
Pyrethrins (0.33%) synergized with piperonyl butoxide (4%), various formulations*	Natural botanicals	Topical application for 10 minutes to dry hair	Allergic reactions in individuals with sensitivity to chrysanthemums, ragweed and related plants	Poor-fair; resistance common
Permethrin cream rinse or lotion (1%)*	Synthetic pyrethroid	Topical application for 10 minutes to clean, dry hair	None	Poor-fair; resistance common
Permethrin cream (5%)†	Synthetic pyrethroid	Topical overnight application to clean, dry hair	Allergic contact dermatitis in individuals with sensitivity to formaldehyde	Poor-fair; resistance common
Lindane shampoo (1%)	Organochlorine	Topical application for 4 minutes to clean, dry hair, then add water to lather and rinse	Potential CNS toxicity; not recommended for infants, children, breastfeeding mothers or pregnant women (category C)	Poor; resistance common
Carbaryl shampoo (0.5%)	Carbamate cholinesterase inhibitor	Topical application for 8–12 hours	Possible carcinogen	Poor-fair; resistance common (not approved in the US)
Benzyl alcohol lotion (5%)§	Alcohol	Topical application for 10 minutes to dry hair	Potential skin irritation	Good; no resistance noted to date
Dimethicone (4%)	Silicone oil	Topical application for 15 minutes or overnight	None	Good
Spinosad cream rinse (0.9%)§	Bacterial fermentation product	Topical application for 10 minutes to dry hair	None	Good; no resistance noted to date
Malathion lotion or gel (0.5%)‡	Organophosphate cholinesterase inhibitor	Topical application for 8–12 hours to dry hair (Ovide® and gel products with isopropyl alcohol base are effective at 20 minutes)	Flammable isopropyl alcohol base; burning or stinging at sites of eroded skin	Excellent (in US); resistance noted in Europe and Australia, but to date not in US
Ivermectin solution (0.5%)§	Avermectin	Topical application for 10 minutes to dry hair	Potential skin and eye irritation	Excellent; no resistance noted to date
Oral ivermectin (available as 3 mg tablets)		Oral dose of 200–400 mcg/kg	Potential CNS toxicity; not recommended for children weighing <33 pounds (15 kg), breastfeeding mothers or pregnant women (category C)	

Mites of dermatological importance

1. *Sarcoptes scabiei*.
2. *Demodex folliculorum*.
3. Cheyletiella mites.
4. House dust mites.
5. Storage mites.

Scabies

- ▶ Def:- pruritic condition caused by sarcoptes scabiei.
- ▶ Epidemiology:- world wide all age.
- ▶ Predisposed by malnutrition, overcrowding & poor hygiene.
- ▶ Aetiopathogenesis:- caused by sarcoptes scabiei var hominis fertilized female[8 legs].
- ▶ Lives in s.c. furrow open to make burrow.
- ▶ Immediate & delayed type hypersensitivity.
- ▶ Ways of transmission:- close physical contact (direct & indirect).
- ▶ Lays 3 eggs /day & hatch in 3-4 days.
- ▶ Life cycle 30 days.



Fig. 84.1 Female scabies mite with eggs and scybala in skin scrapings. Note the mite's flattened, oval body and eight legs.

C/pic

- ▶ I.P:- 2-6 weeks after exposure.
- ▶ Symptoms:- intense pruritis increases at night, hot bathes.
- ▶ Asymptomatic carrier.
- ▶ Signs:- bilateral, symmetrical, flexures, interdigital & webbing.
- ▶ Men → penis & scrotum.
- ▶ Women → areola, nipple & vulva.
- ▶ Infant:- palm, sole , scalp & face.
- ▶ Burrow (1ry lesion) is pathognomonic sign.
- ▶ 2ry lesion → excoriation, dermatitis & 2ry infection.



Fig. 84.3 Scabies. A, B Erythematous papules, linear burrows, areas of crusting and acral vesiculopustules in two infants with scabies. C Close-up of a linear burrow. D, E Penile involvement with erythematous papules and nodules. F Nodular scabies in an infant. A, B, Courtesy, Julie V. Schaffner, MD. E, Courtesy, Robert Hartman, MD.

Complications

- ▶ Eczema
- ▶ 2ry infection.
- ▶ Impetigo (streptococci) → glomerulonephritis.
- ▶ Not vector for any in systemic diseases.

❑ Investigations

- ❑ Clinical diagnosis (criteria) :- night pruritis, lesion distribution & household member.
- ❑ Light microscope
- ❑ Dermoscopy.
- ❑ Wood's light → fluroscent test.
- ❑ Burrow ink test & biopsy.

Dermoscopy of scabies

- ▶ Brown triangular shape jet air-plan + thin tract of whitish scale
 - ❑ Skin biopsy:-
 - ❑ Spongiosis, papillary edema, eosinophils & pink pig tail like structure.
 - ❑ DD
- 1. Dermatitis.
- 2. Arthropod bite.
- 3. DH.
- 4. Bullous Pemphigoid.
- 5. L.C histiocytosis.

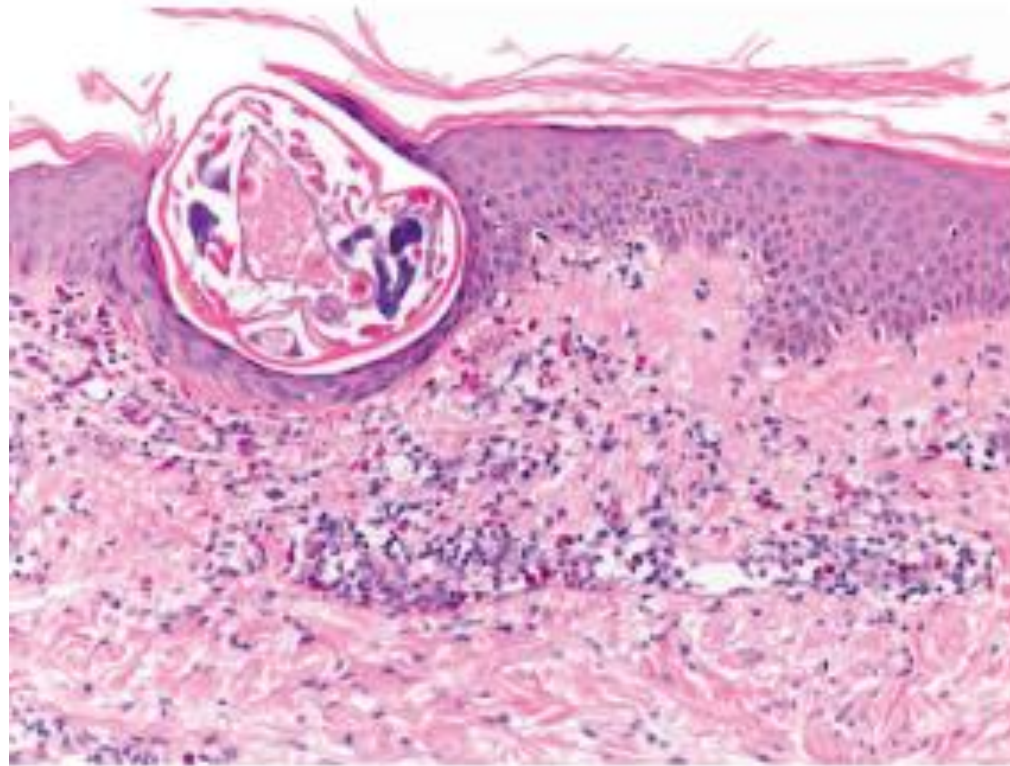


Fig. 84.6 Scabies – histologic features. A mite is evident in the stratum corneum and an eosinophil-rich dermal infiltrate is present. Courtesy, Lorenzo Cerroni, MD.

Treatment of scabies

- General guidelines:-

2 applications 1 wk. apart.

Overnight, entire body.

Wash clothes, linens & towels.

All family members should be treated simultaneously.

Pets NOT harbor human mites.

- Topical ttt

Permethrin 5%, benzyl benzoate & sulphur.

- ▶ Oral

- ▶ Ivermectin 200-400 Mg/kg.

TOPICAL AND ORAL TREATMENTS FOR SCABIES

Therapy	Administration	Concerns	Efficacy & resistance	Use in infants	FDA pregnancy category
Permethrin cream (5%)	Topically overnight on days 1 and 8	Allergic contact dermatitis in individuals with sensitivity to formaldehyde	Good, but some signs of tolerance developing	FDA approved for infants ≥ 2 months of age	B
Lindane lotion or cream (1%)	Topically overnight on days 1 and 8	Potential CNS toxicity, especially in individuals weighing < 110 pounds (50 kg), the elderly and those with crusted scabies, other skin conditions or a history of seizures	Poor, resistance very common	Not recommended for infants, children or breastfeeding mothers; contraindicated in premature infants	C (but not recommended)
Crotamiton lotion or cream (10%)	Topically overnight for 3–5 days	Irritant contact dermatitis, especially in areas of denuded skin	Very poor; has antipruritic properties and may be used for postscabetic pruritus	Not established, but considered safe	C
Sulfur ointment (5–10%)	Topically overnight for 3 successive days	Toxicity studies not performed	Good	Not established, but considered safe	Not rated, but considered safe
Ivermectin (available as 3 mg tablets)	Oral dose of 200–400 mcg/kg on day 1 and repeat on day 8 or 14***	Potential CNS toxicity in infants and young children	Excellent	Safety not established for children weighing < 33 pounds (15 kg) or breastfeeding mothers	C (but generally not recommended for scabies in pregnant women)

Causes of resistance

- ▶ Poor compliance.
- ▶ Resistant scabicide.
- ▶ Reinfection.
- ▶ Psychogenic → acrophobia.
- ▶ Post-scabetic nodule (nodular scabies)

Clinical variants

1. Ordinary.
2. Incognito.
3. Crusted (Norwegian).
4. Animal.
5. Infantile
6. Nodular.
7. In the clean

Crusted (Norwegian)

- ▶ Million of mites.
- ▶ R.F. immunodeficiency, old age, M R, leprosy & paraplegia.
- ▶ Decreased sensation & inability to scratch
- ▶ C/pic:-

Itching, marked hyperkeratosis.

Acral ppk

Lymphadenopathy, eosinophilia.

Increased IG E.

Complications

2ry infection -erythoderma.

DD

Eczema, PS & Darrier.

Ttt

Hospitalization & isolation.

Keratolytic

Oral & topical Scabicial → Ivermectin 3-7 doses



Fig. 84.4 Crusted scabies. Scabies incognito in which a patient with impaired sensory function had an asymptomatic hyperkeratotic hand "rash". *Courtesy, Joyce*

Animal

- ▶ *Sarcoptes scabiei varcanis*.
- ▶ Mites of animal are not source of human infestations.
- ▶ Bite reaction.

- ▶ *C.pic*

Small pruritic wheals / papules like human scabies but not burrow.

In the chest, abdomen, thigh, forearm with self limitation & resolves spontaneously.

Treatment of affected animal should be treated [ectometrin & ivermectin).

Infantile

- ▶ Pruritis.
- ▶ Burrow.
- ▶ Eczema.
- ▶ Impetigo.
- ▶ Vesiculo-acral distribution.
 - ▶ DD→ atopy - acropustular.

No burrows.

No human human transmission.

Self limited.

Changed distribution.

Short I.P.

Urticarial manifestations.

Nodular scabies

- ▶ Hypersensitivity reaction.
- ▶ Intensely pruritic papule, nodule, in groin, axilla & male genitalia.
- ▶ Persistent months after ttt

Cutaneous larva migrans

- ▶ Serpaenous cutaneous eruption caused by animal ancylostoma.
- ▶ Intense pruritis → distal extrimities & buttocks.
- ▶ 2ry infection.
- ▶ Loeffler's syndrome → migratory pulmonary infiltration & eosinophilia.
- ▶ Albendazol - ivermectin.



